

KADI SARVA VISHWAVIDYALAYA
B.E 1st SEMESTER EXAMINATION (DECEMBER 2023)
SUBJECT: Engineering Mathematics-I (New)(Code: CC101-N)

Date: 28/12/2023

Time: 3 hour

Marks: 70

Instruction:

1. Answer each section in separate Answer Sheet.
2. Use of scientific Calculator is permitted.
3. All questions are compulsory.
4. Indicate **clearly** the option you attempted along with its respective question number.
5. Use the last page of main supplementary for rough work.

Section:1

Q.1 (a) Use L'Hospital's rule to evaluate $\lim_{x \rightarrow 0} \frac{2x - x \cos(x) - \sin(x)}{2x^3}$. [05]

(b) Evaluate $\int_0^1 \int_x^{x^2} (xy) \, dx \, dy$. [05]

(c) Find the Jacobian $J = \frac{\partial(u, v)}{\partial(x, y)}$ for $u = x \sin(y)$ and $v = y \sin(x)$. [05]

OR

(c) Examine the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{n^3+1}$. [05]

Q.2 (a) If $u = x^2 y^3$, where $x = \log(t)$, $y = e^t$ then find $\frac{du}{dt}$. [05]

(b) Find the equation of tangent plane and normal line at point $p(1, 2, 4)$ on the surface $x^2 + y^2 + z = 9$. [05]

OR

Q.2 (a) Find the area between the parabola $y^2 = 4ax$ and line $x^2 = 4ay$. [05]

(b) Trace the curve $y^2(a - x) = x^3$, $a > 0$. [05]

Q.3 (a) If $u = \tan^{-1} \left(\frac{x^4 + y^4}{x^2 + y^2} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin(2u)$. [05]

(b) If $u = f(x - y, y - z, z - x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. [05]

OR

Q.3 (a) Determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{2^n}{n^3+1}$ by D'Alembert's ratio test. [05]

(b) Using reduction formulae evaluate integral $\int_0^{\infty} \frac{1}{(1+x^2)^6} \, dx$. [05]

Section:2

Q.4 (a) Expand $f(x) = x^5 - x^4 + x^3 - x^2 - x - 1$ in powers of $(x - 1)$ by using Taylor's series. [05]

(b) If $u = e^{xyz}$ then show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2 y^2 z^2) e^{xyz}$. [05]

(c) Evaluate $\iiint_E 2x \, dz dy dx$, where E is the region under the plane $2x + 3y + z = 6$ that lies in the first octant. [05]

OR

(c) Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ [05]

Q.5 (a) If $u = \log(x^2 + y^2)$ then verify that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$. [05]

(b) Evaluate $\iint_R xy \, dx \, dy$, where R is the region bounded by x -axis, line $x = 2a$ and the curve $x^2 = 4ay$. [05]

OR

Q.5 (a) Examine the convergence of the series $\sum_{n=1}^{\infty} \left[\frac{1}{(1 + \frac{1}{n})^{n^2}} \right]$ by using Cauchy's root test. [05]

(b) Trace the curve $r = 2a \cos \theta$. [05]

Q.6 (a) Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$ by changing order of integration. [05]

(b) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x+y+z)^2}$ [05]

OR

Q.6 (a) Evaluate $\int_0^2 \int_2^4 \int_0^3 x^2 y z^2 \, dy dz dx$. [05]

(b) Discuss the maxima and minima of $x^2 + y^2 + 6x + 12$. [05]

❖❖❖❖❖❖